DIVYA RANA

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RESEARCH PROFILE

Observational cosmologist using weak lensing and galaxy clustering to connect galaxies to their dark matter halos across scales, from the stellar initial mass function (IMF) to halo outskirts and to cosmology with cosmic shear and gravitational-wave sirens. I combine Euclid, HSC and eROSITA data with image simulations, and coordinate galaxy–galaxy lensing for **Euclid Data Release 1**.

RESEARCH INTERESTS

- Large-scale structure and cosmology
- Galaxy–halo connection: lensing & clustering
- Stellar IMF and galaxy mass budgets
- Weak-lensing image simulations & calibration
- Halo outskirts, splashback and environment
- Gravitational-wave cosmology, tests of ΛCDM

RESEARCH HIGHLIGHTS

- **Small-scale lensing as an IMF probe for Euclid:** image simulations show lens light biases small-scale galaxy–galaxy lensing, which Galfit subtraction cuts ∼10×; forecast ∼8.5% precision on the IMF mismatch parameter, distinguishing Chabrier, Kroupa and Salpeter IMFs (arXiv:2610.01541).
- **Independent photo-z calibration for HSC Y3 cosmic shear:** blinded shear-ratio analysis with GAMA, SDSS and DESI lenses; joint $S_8 = 0.773 \pm 0.031$ (PRD 2026).
- **Edges of dark matter halos:** best splashback constraints for X-ray clusters at the time (eFEDS, MNRAS 2023); mean cluster mass-accretion rate from splashback (PRD 2026, student-led).
- **Galaxy–halo connection:** 5% constraint on the group halo mass–luminosity relation (GAMA×HSC, MNRAS 2022); first upper limit on orphan-galaxy fractions (MNRAS 2022, student-led).
- **Gravitational-wave cosmology:** first simulation-based demonstration that GW–galaxy cross-correlations constrain H_0 to <10% with 50 BBHs even with no hosts in the catalogue (ApJ 2020).

APPOINTMENTS

Postdoctoral Researcher, Leiden Observatory, Leiden University *Oct 2023 – Present*
Supervisor: Prof. Dr. Henk Hoekstra

EDUCATION

PhD in Physics, Inter-University Centre for Astronomy and Astrophysics, India *2019 – 2023*
Thesis: *Probing Cosmology with Large Scale Structure Correlations*; Advisor: Prof. Surhud S. More
Junior Research Fellow, IUCAA Graduate School Program *2018 – 2019*

Integrated MSc in Physics, National Inst. of Science Education and Research, India *2013 – 2018*
Thesis: *Similarities between freely cooling granular gases and LSS formation under gravity*;
Advisors: Prof. Nishikanta Khandai, Prof. Anamitra Mukherjee

FELLOWSHIPS, GRANTS AND AWARDS

- **2025:** Leids Kerkhoven-Bosscha Fonds grant for an invited talk at Shanghai Jiao Tong University.
- **2024:** Erasmus+ grant to deliver a lecture series at Universidad de Antioquia, Colombia.
- **2020:** Selected for the Program for Astrophysics Visitor Exchange at Stanford (PAVES).
- **2018:** All India Rank 17 (JEST) and 100 (CSIR-NET JRF); PhD offers from IISc, NCRA–TIFR, IISER Mohali.
- **2013–2018:** INSPIRE Scholar, Department of Science and Technology, Government of India.

RESEARCH LEADERSHIP AND COLLABORATIONS

Key Project Coordinator, Galaxy–Galaxy Lensing, Euclid Data Release 1 *2025 – Present*

- Coordinate the multi-institute, cross-working-group delivery of the DR1 galaxy–galaxy lensing science products (timelines, milestones, communication); liaison to Euclid Consortium management.

Memberships: Euclid Consortium; HSC Collaboration; LIGO Scientific Collaboration; eROSITA (external collaborator).

SUPERVISION AND MENTORING

- **PhD co-supervision:** Gimena Martín Girardi (Leiden, since 2024); environment-dependent halo model with FLAMINGO, paper in preparation.
- **LEAPS summer students:** Leonid Sajkov (2024; now PhD student, Stanford University); Dannaé Díaz-Axtle (2025; now PhD student, Durham University; first-author paper arXiv:2609.15416); Yu Ting (2026).
- **Undergraduates (Leiden University):** Sanne Bosch (2024–2025); Milan Tol (2025–2026); Hugo Bogaart (since 2026).
- **IUCAA PhD mentees:** Jitendra Joshi (PRD 2026; arXiv:2609.15375); Amit Kumar (MNRAS 2022).

TEACHING

- **Co-lecturer**, *Weak Gravitational Lensing* course, Leiden University (2026).
- **Invited lecturer**, one-week weak-lensing course with tutorials, Astronomy Twinning Program (Leiden Observatory–Univ. de Antioquia), Medellín, Colombia (2024; Erasmus+ funded).

SELECTED TALKS

- **Plenary:** Clustering the Cosmos: From Galaxy Formation to the Expansion of the Universe, *NAC 2026*, Netherlands (2026).
- **Invited:** Expanding the boundaries of dark matter halos, Shanghai Jiao Tong University, China (2025).
- **Invited:** Expansion rate from GW–galaxy cross-correlations, *21st Century Cosmology*, Ashoka Univ., India (2024).
- Galaxy–halo connection using the HSC survey, *Euclid WL SWG Meeting*, Innsbruck, Austria (2024).
- Seminars at Kavli IPMU and ICRR, Tokyo (2022): HSC halo boundaries; invisible sirens.
- Weak lensing of GAMA galaxy groups with Subaru HSC, *ICGC*, IISER Mohali, India (2020).

SERVICE AND OUTREACH

- Referee for *The Astrophysical Journal* and *Astronomy & Astrophysics*.
- Organizer, Leiden/ESA Astrophysics Program for Summer Students (LEAPS).
- Research featured in *Nature India* (March 2026).
- Member, Science Popularization Committee, IUCAA.

COMPUTING SKILLS

Python (scientific stack, GalSim), C, C++, Mathematica; SExtractor, Galfit; LaTeX; Linux and macOS.

REFERENCES

Prof. Dr. Henk Hoekstra	Leiden University, The Netherlands	hoekstra@strw.leidenuniv.nl
Prof. Dr. Konrad Kuijken	Leiden University, The Netherlands	kuijken@strw.leidenuniv.nl
Prof. Surhud S. More	IUCAA, Pune, India	surhud@iucaa.in
Prof. Hironao Miyatake	KMI, Nagoya University, Japan	miyatake@kmi.nagoya-u.ac.jp
Prof. Sukanta Bose	Washington State University, USA	sukanta@wsu.edu

LIST OF PUBLICATIONS

10 significant contribution papers (4 first-author; 4 led by students I supervised or mentored; 7 published, 3 under review), plus LIGO–Virgo–KAGRA collaboration papers. Full list on ADS.
Bold: D. Rana; †: student supervised or mentored by D. Rana.

Significant Contribution Papers

- **D. Rana**, H. Hoekstra, K. Kuijken, A. Sonnenfeld, M. Tajalli (2026). Weighing Galaxies Inside-Out: Small-Scale Lensing and the Stellar Mass Problem. *Under review*, arXiv:2610.01541
- D. Díaz-Axtle†, P. Awad, **D. Rana**, H. Hoekstra (2026). Influence of the large-scale structure velocity field on halo mass assembly within cosmic filaments. *Under review*, arXiv:2609.15416
- J. Joshi†, **D. Rana**, S. More (2026). Assembly Bias in eRASS1 X-ray Galaxy Clusters: Accounting for Non-uniform Survey Exposure. *Under review*, arXiv:2609.15375
- **D. Rana**, S. More, H. Miyatake, S. Sugiyama, T. Zhang, M. Shirasaki (2026). Hyper Suprime-Cam Y3 results: Photo-z bias calibration with lensing shear ratios and cosmological constraints from cosmic shear. *Physical Review D*, 113, 063534. arXiv:2508.21681
- J. Joshi†, **D. Rana**, S. More, M. Klein (2026). Splashback radius and the mass accretion rate of RASS MCMF galaxy clusters. *Physical Review D*, 113, 023512. arXiv:2506.08925
- **D. Rana**, S. More, H. Miyatake, et al. (2023). The eROSITA Final Equatorial-Depth Survey (eFEDS): Splashback radius of X-ray galaxy clusters using galaxies from HSC survey. *MNRAS*, 522, 4181. arXiv:2301.03626
- P. K. Mishra, **D. Rana**, S. More (2023). Stellar mass dependence of galaxy size–dark matter halo radius relation probed by Subaru-HSC survey weak lensing measurements. *MNRAS*, 526, 2403. arXiv:2301.04664
- **D. Rana**, S. More, H. Miyatake, T. Nishimichi, M. Takada, A. S. G. Robotham, A. M. Hopkins, B. W. Holwerda (2022). The Subaru HSC weak lensing mass-observable scaling relations of spectroscopic galaxy groups from the GAMA survey. *MNRAS*, 510, 5408. arXiv:2107.05641
- A. Kumar†, S. More, **D. Rana** (2022). Subaru HSC weak lensing of SDSS redMaPPer cluster satellite galaxies: Empirical upper limit on orphan fractions. *MNRAS*, 517, 4389 (erratum: MNRAS, 531, 2024). arXiv:2205.00018
- S. Bera, **D. Rana**, S. More, S. Bose (2020). Incompleteness Matters Not: Inference of H_0 from BBH–galaxy cross-correlations. *The Astrophysical Journal*, 902, 79. arXiv:2007.04271

In Preparation

- **D. Rana**, et al. Cosmology at the Edge: The Physical Imprint of the Halo Periphery.
- G. Martín Girardi†, **D. Rana**, H. Hoekstra. Environmental dependence in the halo model informed by FLAMINGO simulations. To be submitted.

Selected LIGO–Virgo–KAGRA Collaboration Publications

Co-author as a member of the LIGO Scientific Collaboration; selected recent papers listed.

- LVK Collaboration (incl. **D. Rana**) (2026). GWTC-4.0: Population Properties of Merging Compact Binaries. *ApJL*, 1005, L51. arXiv:2508.18083
- LVK Collaboration (incl. **D. Rana**) (2026). GWTC-4.0: Methods for Identifying and Characterizing Gravitational-wave Transients. *ApJL*, 1004, L21. arXiv:2508.18081
- LVK Collaboration (incl. **D. Rana**) (2025). GWTC-4.0: An Introduction to Version 4.0 of the Gravitational-Wave Transient Catalog. *ApJL*, 995, L18. arXiv:2508.18080
- LVK Collaboration (incl. **D. Rana**) (2025). Search for Gravitational Waves Emitted from SN 2023ixf. *ApJ*, 985, 183. arXiv:2410.16565
- LVK Collaboration (incl. **D. Rana**) (2025). Search for Continuous Gravitational Waves from Known Pulsars in the First Part of the Fourth LIGO–Virgo–KAGRA Observing Run. *ApJ*, 983, 99. arXiv:2501.01495

- G. Raman, et al., & LVK Collaboration (incl. **D. Rana**) (2025). Swift-BAT GUANO Follow-up of Gravitational-wave Triggers in the Third LIGO–Virgo–KAGRA Observing Run. *ApJ*, 980, 207. arXiv:2407.12867
- LVK Collaboration (incl. **D. Rana**) (2024). Search for Eccentric Black Hole Coalescences during the Third Observing Run of LIGO and Virgo. *ApJ*, 973, 132. arXiv:2308.03822
- LVK Collaboration (incl. **D. Rana**) (2024). Observation of Gravitational Waves from the Coalescence of a 2.5–4.5 $M_\odot$ Compact Object and a Neutron Star. *ApJL*, 970, L34. arXiv:2404.04248
- LVK Collaboration (incl. **D. Rana**) (2023). Search for subsolar-mass black hole binaries in the second part of Advanced LIGO's and Advanced Virgo's third observing run. *MNRAS*, 524, 5984. arXiv:2212.01477
- LVK Collaboration (incl. **D. Rana**) (2023). Open Data from the Third Observing Run of LIGO, Virgo, KAGRA, and GEO. *ApJS*, 267, 29. arXiv:2302.03676
- LVK Collaboration (incl. **D. Rana**) (2022). Model-based Cross-correlation Search for Gravitational Waves from the Low-mass X-Ray Binary Scorpius X-1 in LIGO O3 Data. *ApJL*, 941, L30. arXiv:2209.02863